

Prompt-driven CAD-grounded ROI and Viewpoint Planning for Robotic Visual Inspection

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Online Resource 1. This file provides supplementary implementation details for the proposed CAD-grounded inspection-planning framework. It documents the CAD perception schema, rule-based semantic face labels, prompt interpretation, LLM-based CAD-tool reasoning, measurement-point and viewpoint generation, and CAD-level synthetic defect generation. These details support reproducibility of the implementation while keeping the main manuscript focused on the perception–reasoning–action interface and the experimental findings.

S1 Supplementary perception schema and geometric descriptors

This supplementary section documents the implementation-level details of the perception stage. The perception stage is an offline CAD-and-language interpretation step. It does not perform physical sensing. Its role is to convert the input STEP model and natural-language inspection request into structured representations used by the reasoning and action stages.

S1.1 CAD entity schema

The perception module converts the input STEP model into a structured CAD representation that can be queried by the reasoning module and reused by the deterministic planning stage. The schema consists of five entity groups:

- **Global model:** model-scale attributes, including bounding box, characteristic scale, surface area, volume, face/edge counts, and detected hole-like entity count.
- **Face:** surface-level attributes, including surface type, area, centroid, normal, curvature summaries, boundary edges, adjacent faces, and optional semantic labels.
- **Edge:** boundary-level attributes, including curve type, length, tangent, endpoints, incident faces, dihedral angle, and boundary type.
- **Hole-like entity:** compact hole-related attributes, including center, radius, axis, depth, wall faces, rim edges, and entry/exit support faces.
- **Patch:** aggregated surface attributes, including member faces, patch area, centroid, normal, semantic type, boundary edges, and neighboring patches.

Global model attributes provide scale normalization and workspace context. Faces and edges form the basic geometric and topological units for ROI grounding, while hole-like entities provide prompt-aligned abstractions for hole-, bore-, passage-, and rim-related inspection tasks. Patches serve as intermediate planning units that connect language-conditioned ROI reasoning with deterministic measurement-point and viewpoint generation.

S1.2 Geometric descriptor computation

The underlying B-Rep solid is denoted by

$$\mathcal{S} = (\mathcal{F}, \mathcal{E}, \mathcal{H}, \mathcal{B}), \quad (\text{S1})$$

where $\mathcal{F} = \{f_i\}_{i=1}^{N_F}$, $\mathcal{E} = \{e_j\}_{j=1}^{N_E}$, $\mathcal{H} = \{h_k\}_{k=1}^{N_H}$ denote the face, edge, and hole-like entity sets, respectively, and $\mathcal{B}$ denotes boundary-loop and body-level topological information.

For each face f_i , the parser treats the surface as a parametric map

$$\mathbf{r}_i : \Omega_i \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}^3, \quad (\text{S2})$$

and computes the area, centroid, and representative normal by

$$A_i = \iint_{\Omega_i} \left\| \frac{\partial \mathbf{r}_i}{\partial u} \times \frac{\partial \mathbf{r}_i}{\partial v} \right\| du dv, \quad (\text{S3})$$

$$\mathbf{c}_i = \frac{1}{A_i} \iint_{\Omega_i} \mathbf{r}_i(u, v) \|\mathbf{r}_{i,u} \times \mathbf{r}_{i,v}\| du dv, \quad (\text{S4})$$

$$\mathbf{n}_i = \frac{\mathbf{r}_{i,u}(\bar{u}_i, \bar{v}_i) \times \mathbf{r}_{i,v}(\bar{u}_i, \bar{v}_i)}{\|\mathbf{r}_{i,u}(\bar{u}_i, \bar{v}_i) \times \mathbf{r}_{i,v}(\bar{u}_i, \bar{v}_i)\|}. \quad (\text{S5})$$

Curvature cues are aggregated over a regular UV sample set $\mathcal{U}_i$ as

$$\bar{H}_i = \frac{1}{|\mathcal{U}_i|} \sum_{(u,v) \in \mathcal{U}_i} |H_i(u, v)|, \quad \bar{K}_i = \frac{1}{|\mathcal{U}_i|} \sum_{(u,v) \in \mathcal{U}_i} |K_i(u, v)|, \quad (\text{S6})$$

which provide compact differential-geometric summaries for identifying nearly planar, single-curvature, and free-form regions.

Each edge e_j is represented as a parametric curve $\gamma_j : [a_j, b_j] \rightarrow \mathbb{R}^3$ with length

$$L_j = \int_{a_j}^{b_j} \|\gamma'_j(t)\| dt, \quad (\text{S7})$$

and a representative tangent

$$\mathbf{t}_j = \frac{\gamma'_j(t_j^*)}{\|\gamma'_j(t_j^*)\|}. \quad (\text{S8})$$

Hole-like entities are summarized as $h_k = (\mathbf{c}_k, r_k, d_k, \mathbf{a}_k)$, where $\mathbf{c}_k$ is the center, r_k the radius, d_k the approximate depth, and $\mathbf{a}_k$ the axis direction. When cylindrical side-wall evidence is available, the depth is approximated by

$$d_k \approx \frac{A_k^{\text{lat}}}{2\pi r_k}. \quad (\text{S9})$$

S1.3 Topology relation construction

The topology graph is represented as

$$\mathcal{G}_{\text{topo}} = (V, E_{\text{topo}}), \quad E_{\text{topo}} = E_{FE} \cup E_{FF} \cup E_{FH}. \quad (\text{S10})$$

A face–edge relation is inserted when an edge belongs to the boundary loop of a face. A face–face relation is inserted when two faces share at least one topological edge. A face–hole relation is inserted when a face is associated with a hole-like entity as a wall, rim, entry support, or exit support.

For adjacent faces, the representative normal angle is computed as

$$\theta_{ij} = \cos^{-1}(\text{clip}(\mathbf{n}_i^\top \mathbf{n}_j, -1, 1)). \quad (\text{S11})$$

The parser stores the shared edge ID, normal angle, and local transition type for each adjacent face pair.

S1.4 Patch decomposition

The patch graph is constructed by retaining adjacent face pairs that are both geometrically smooth and semantically compatible:

$$(f_i, f_j) \in E_p \iff (f_i, f_j) \in E_{FF} \wedge \theta_{ij} < \tau_p \wedge \text{comp}(\ell_i, \ell_j) = 1. \quad (\text{S12})$$

Connected components of this graph define the patch set

$$\mathcal{P} = \{P_m\}_{m=1}^M. \quad (\text{S13})$$

Patch-level aggregate descriptors are computed as

$$A(P_m) = \sum_{f_i \in P_m} A_i, \quad \mathbf{c}_m = \frac{1}{A(P_m)} \sum_{f_i \in P_m} A_i \mathbf{c}_i, \quad (\text{S14})$$

$$\mathbf{n}_m = \frac{\sum_{f_i \in P_m} A_i \mathbf{n}_i}{\left\| \sum_{f_i \in P_m} A_i \mathbf{n}_i \right\|}. \quad (\text{S15})$$

Patch decomposition intentionally preserves inspection-relevant transition surfaces such as chamfers, fillets, rims, and hole boundaries instead of merging them into large adjacent host faces.

S2 Rule-based semantic face labels

The CAD perception module assigns lightweight semantic labels to faces using rule-based geometric and topological cues. These labels are not evaluation targets; they are intermediate descriptors used to guide ROI grounding, patch construction, and inspection planning.

The labels are organized into four groups:

- **Support-surface labels.** These labels identify broad reference regions.
 - *Planar support*: a large planar face that can serve as a datum, host, mounting, sealing, or support surface.
 - *Outer planar*: a large exterior planar face used for outer-surface or full-surface inspection.
- **Cylindrical and hole-related labels.** These labels capture rounded or opening-related structures.
 - *Cylindrical wall*: a cylindrical face with dominant one-directional curvature, including bores, passages, pins, and rounded walls.
 - *Hole wall*: a cylindrical wall connected to circular boundary loops.
 - *Rim/opening*: a narrow band around a hole-like opening, used for burr, rim, exit-edge, and boundary-sensitive inspection.
- **Transition labels.** These labels preserve small but inspection-relevant boundary regions.
 - *Chamfer*: a narrow planar or conical transition face.
 - *Fillet*: a smooth curved transition between parent faces.
 - *Small transition*: a small face adjacent to geometrically dissimilar faces, preserved separately during patch construction.
- **Recess and wall labels.** These labels describe local interior or side-facing regions.
 - *Pocket floor*: a planar face enclosed by higher neighboring walls or inward boundary structure.
 - *Side wall*: a side-facing planar or weakly curved face used for wall, rib-side, deformation, or bulge inspection.

S3 Prompt interpretation and CAD-exploration-level classification

The language-side perception module converts the natural-language inspection request q into a compact task representation:

$$\phi = (\lambda, \tau, \omega, \Gamma^{\mathcal{L}}). \quad (\text{S16})$$

This representation provides an initial task hypothesis for the reasoning module. It describes what the user appears to request, but it does not include evaluation-only target labels, face IDs, or Target–Datum–Envelope (TDE) construction rules.

The prompt-side schema consists of four fields:

- **CAD-exploration level λ .** Indicates the expected reasoning depth of the request, such as shallow CAD exploration, standard CAD exploration, or deep CAD exploration. This field controls the available tool families, reasoning depth, and function-call budget.
- **Target cues τ .** Captures linguistic hints about inspection-relevant CAD entities, such as holes, rims, edges, fillets, chamfers, pockets, slots, sealing surfaces, or mounting interfaces. These cues guide CAD-tool search and feature-family prioritization.
- **Objective and density ω .** Describes the intended inspection objective, such as full inspection, burr inspection, surface-defect inspection, dimensional checking, or dense local inspection. This field determines whether the planner should favor broad coverage, boundary-sensitive sampling, or denser local sampling.
- **Language-derived sensing constraints $\Gamma^{\mathcal{L}}$.** Stores soft sensing requirements inferred from the prompt, such as small defects, fine scratches, large deformation, close inspection, or multi-view inspection. These constraints are later specialized into ROI-level sensing requirements by the reasoning and planning stages.

Together, these fields separate language interpretation from geometric grounding. The prompt parser only extracts the user’s apparent intent, while the subsequent CAD-grounded reasoning stage determines the actual ROI, sampling strategy, and sensing constraints.

S4 LLM reasoning configuration and CAD-tool inventory

S4.1 LLM reasoning configuration

The reasoning module uses an LLM as a function-calling controller over a fixed CAD-analysis interface. The LLM selects tools, interprets structured CAD observations, and synthesizes an ROI strategy. It does not directly generate measurement points, viewpoints, visibility values, or evaluation metrics.

- **Model.** Qwen3-Coder-30B-A3B-Instruct
- **Input context.** The model receives the structured prompt representation ϕ , the CAD summary, available tool definitions, the CAD-exploration level configuration, and accumulated tool-call observations $\mathcal{Z}_{1:t}$.
- **Tool-calling mode.** The model operates over a fixed function-calling interface. It can only call predefined CAD-analysis tools and receives structured observations returned by those tools.
- **Decoding parameters.** The decoding parameters are set to `temperature=0`, `top_p=1`, `top_k=1`, `seed=42`, and `num_predict=4096`.
- **CAD-exploration-level budget.** Shallow CAD exploration (λ_1), standard CAD exploration (λ_2), and deep CAD exploration (λ_3) use maximum tool-call budgets of 6, 12, and 16, respectively, under a global iteration cap of 16.
- **Output format.** The model outputs a structured ROI strategy containing patch IDs, face IDs, priority, density, multi-view flag, and ROI-level sensing constraints.
- **Validation policy.** Invalid patch IDs, face IDs, unsupported density labels, or out-of-range priority values are rejected or normalized before deterministic execution.

- **Leakage prevention.** Evaluation-only target labels, ground-truth face IDs, TDE component labels, and TDE construction rules are not included in the model input.

S4.2 CAD-exploration-level-dependent CAD-tool access

Online Table 1. CAD-exploration-level-dependent CAD analysis tools and agent-loop controls. Shallow CAD exploration (λ_1), standard CAD exploration (λ_2), and deep CAD exploration (λ_3) access tools 1–6, 1–12, and 1–16, respectively, with matching call budgets of 6/12/16 under a global iteration cap of 16. The CAD-exploration level controls tool availability and reasoning depth; it is not an evaluation prompt category.

ID	Tool	Role
1	CAD summary	Provides global model statistics and scale context.
2	Feature classifier	Groups entities into manufacturing feature families.
3	Importance scoring	Estimates face-level inspection relevance.
4	Property ranking	Selects top- N entities by geometric property.
5	Patch expansion	Expands selected faces into patch context.
6	Patch summary	Summarizes patch geometry and face membership.
7	Feature filtering	Filters entities by geometric or topological rules.
8	Feature details	Retrieves attributes of selected entities.
9	Patch boundary	Identifies boundary, interior, and opening faces.
10	Spatial filtering	Selects entities by relative model position.
11	Family ranking	Ranks instances within a feature family.
12	Isolation detection	Finds topologically isolated feature instances.
13	Patch neighbors	Retrieves adjacent patches and relationships.
14	Description match	Matches language descriptions to candidate faces.
15	Spatial clustering	Clusters entities for group-level reasoning.
16	Adjacency search	Finds neighbors from selected source entities.

The CAD-exploration level λ assigned by the prompt-side perception module configures both the available CAD-tool families and the maximum number of function calls. Shallow CAD exploration uses only global summary, feature-ranking, and patch-expansion tools; standard CAD exploration additionally enables feature filtering, feature details, boundary analysis, and spatial filtering; deep CAD exploration further enables neighborhood expansion, description-based search, spatial clustering, and adjacency search. Online Table 1 summarizes the resulting tool inventory and call budgets.

S5 Supplementary execution details for measurement-point and viewpoint generation

This supplementary section documents the implementation constants and lower-level formulas for measurement-point (MP) and viewpoint (VP) generation that were omitted from the main text to keep the action stage concise.

S5.1 Measurement-point allocation details

For each ROI group $r_g = (\mathcal{P}_g, \mathcal{F}_g, \pi_g, \rho_g, \chi_g^{\text{MV}}, \Gamma_g)$ the preliminary group budget is computed as

$$b_g = |\mathcal{F}_g| c(\rho_g) \pi_g, \quad (\text{S17})$$

where $c(\text{high}) = 15$, $c(\text{medium}) = 8$, and $c(\text{low}) = 3$ in the current implementation. The total budget is clipped to $[B_{\min}, B_{\max}]$ and partitioned into ROI and non-ROI portions by

$$B_{\text{ROI}} = \alpha_{\text{ROI}} B, \quad B_{\text{non-ROI}} = (1 - \alpha_{\text{ROI}}) B, \quad (\text{S18})$$

with $\alpha_{\text{ROI}} = 0.8$.

At the patch level, density weights are set to $w_\rho(\text{high}) = 3.0$, $w_\rho(\text{medium}) = 1.5$, and $w_\rho(\text{low}) = 1.0$. The current implementation uses $n_{\min} = 3$ MPs per patch before redistributing the remaining budget

proportionally to weighted area. For boundary-sensitive patches, the Poisson-disk sampling policy uses an approximate 70/20/10 split across interior, boundary-band, and opening-rim zones; interior-only patches reassign the boundary-band quota to the interior zone.

After patch-level sampling, point-level ROI membership is optionally refined using anchor points derived from ROI clusters. Let $\{\mathbf{a}_r\}$ denote the anchor set and R_{ROI} the adaptive ROI radius. Each generated MP at $\mathbf{x}_i$ is re-labeled by

$$\chi_{\text{ROI}}(i) = \begin{cases} 1, & \min_r \|\mathbf{x}_i - \mathbf{a}_r\| \leq R_{\text{ROI}}, \\ 0, & \text{otherwise.} \end{cases} \quad (\text{S19})$$

S5.2 Viewpoint-generation defaults

The current default camera configuration is chosen to emulate an Intel RealSense D435-like sensor with

$$(\theta_h, \theta_v, d_w, \eta, s_{\text{obj}}) = (87^\circ, 58^\circ, 150 \text{ mm}, 0.8, 5 \text{ mm}). \quad (\text{S20})$$

Here, θ_h and θ_v denote the horizontal and vertical fields of view, d_w is the nominal working distance, η is the fill factor, and s_{obj} is the nominal object-space support scale used for viewpoint qualification. Unless the prompt overrides the sensing setup, these values are used as the default sensing configuration.

Initial VP candidates are generated from three complementary pose families. Patch-level viewpoints include one frontal macro viewpoint at the patch centroid, eight oblique viewpoints obtained by applying 15° and 30° tilts over four azimuth directions, elongated-patch viewpoints offset along the major patch axis when the aspect ratio exceeds 2, and opening-directed oblique viewpoints tilted by 20° toward local hole directions. Face-level viewpoints include one normal-aligned macro viewpoint and four 15° tilted viewpoints per face. Hole-related cylindrical surfaces receive two axis-aligned viewpoints and eight rim viewpoints generated with a 65° off-axis tilt over four azimuth directions from both axial sides.

Candidate poses are retained only if they remain on the valid viewing side of the surface, remain farther than $d_{\text{min}}^{\text{safe}} = 20 \text{ mm}$ from the nearest MP, and stay within a global model-scale bound. During optimization, a virtual field of view is used:

$$\theta_h^{\text{virt}} = \gamma \theta_h, \quad \theta_v^{\text{virt}} = \gamma \theta_v, \quad (\text{S21})$$

with $\gamma = 0.8$. This narrows each candidate’s effective support and helps prevent overly optimistic single-view coverage estimates.

For every visible VP–MP pair, the composite quality score uses

$$\kappa_{j,i} = 0.65 \kappa_{j,i}^{\text{ang}} + 0.15 \kappa_{j,i}^{\text{dist}} + 0.20 \kappa_{j,i}^{\text{gsd}}. \quad (\text{S22})$$

The greedy set-cover stage is iteratively refined for up to three rounds by regenerating supplemental candidates around uncovered MPs.

S6 Detailed NURBS control-point defect generation

The placement rules described in the main text determine where each synthetic defect is inserted, whereas the displacement models in this section determine how the local CAD surface is geometrically modified after its support region is selected. In Online Table 2, $\mathbf{P}_{ij}$ denotes a NURBS or B-spline control point and $\Delta \mathbf{P}_{ij}$ denotes the displacement applied to that control point. The vector $\hat{\mathbf{n}}_{\text{out}}$ is the outward surface normal, $\hat{\mathbf{a}}_h$ is the detected hole-axis direction, $\hat{\mathbf{r}}$ is the radial direction from the hole center to the rim, and $\hat{\mathbf{t}}_{\text{rim}}$ is the local tangential direction along the rim. The scalar weights W , w_{rim} , w_{patch} , and w_{tear} control local support, rim proximity, angular patch coverage, and sparse tear-like protrusion effects, respectively.

Online Table 2. CAD-level defect displacement models

Defect	Support / weight	Displacement model
Dent	Elliptical UV support with Gaussian falloff, $W = \exp(-\xi^2/2\sigma^2)$ for $\xi < 1$.	$\Delta \mathbf{P}_{ij} = -hW\hat{\mathbf{n}}_{\text{out}}$. This creates a smooth inward depression.
Bulge	Same elliptical Gaussian support as dent.	$\Delta \mathbf{P}_{ij} = +hW\hat{\mathbf{n}}_{\text{out}}$. This creates a smooth outward protrusion.
Scratch	Line-segment support in scaled UV space. The normalized transverse distance is $\xi = d^\perp/W_{\text{half}}$.	$\Delta \mathbf{P}_{ij} = -DB(\xi)R_{ij}\hat{\mathbf{n}}_{\text{out}}$, where B is a V- or U-shaped cross-section and $R_{ij} = 1 + \alpha_{\text{noise}}N_{ij}$ adds mild Perlin-noise roughness.
Burr	Asymmetric rim-distance weight around a detected circular hole, combined with angular Perlin masks for partial coverage and tearing.	$\Delta \mathbf{P}_{ij} = Hw_{\text{rim}}w_{\text{patch}}(\hat{\mathbf{a}}_h + \lambda_{\text{flare}}\hat{\mathbf{r}} + \lambda_{\text{curl}}\hat{\mathbf{t}}_{\text{rim}}) + H\lambda_{\text{tear}}w_{\text{tear}}w_{\text{rim}}\hat{\mathbf{a}}_h$. This models uplift, flare, curl, and sparse tear-like protrusions.

Online Table 3. Hyperparameters for CAD-level synthetic defect generation

Defect	Hyperparameters
Common	Target control points per direction: 30; fixed boundary margin: outer 2 rows and columns.
Dent	Depth $h = 0.3\text{--}0.7$ mm; support radii $r_u, r_v = 3\text{--}12$ mm; Gaussian width $\sigma = 0.45$; noise: none.
Bulge	Height $h = 0.25\text{--}0.6$ mm; support radii $r_u, r_v = 5\text{--}18$ mm; Gaussian width $\sigma = 0.45$; noise: none.
Scratch	Depth $D = 0.08\text{--}0.22$ mm; half-width $W_{\text{half}} = 0.4\text{--}0.8$ mm; profile: V or U; noise amplitude $\alpha_{\text{noise}} = 0.4$; Perlin octaves: 3.
Burr	Height $H = 0.15\text{--}0.8$ mm; rim widths $W_{\text{out}}, W_{\text{in}}$: hole-radius dependent; $k_{\text{out}}, k_{\text{in}} = 2.0, 2.2$; patch coverage 0.55; $\gamma_{\text{tear}} = 2.5$; $\lambda_{\text{flare}}, \lambda_{\text{curl}}, \lambda_{\text{tear}} = 0.30, 0.18, 0.35$.

All synthetic defects are generated at the CAD level by directly editing the control points of NURBS/B-spline surfaces, rather than perturbing a tessellated mesh after geometry export. This design preserves a parametric representation of the modified part and keeps the generated defects compatible with subsequent STEP/B-Rep parsing, ROI grounding, and viewpoint-planning evaluation. Although all defect types share the same control-point editing pipeline, they differ in how the affected region is defined and how the displacement vector is assigned. Dents and bulges use compact elliptical supports in the surface UV domain and move the selected control points along the outward surface normal with opposite signs. Scratches are modeled as narrow line-supported grooves with a V- or U-shaped cross-section and mild procedural roughness generated using Perlin-noise perturbations. Burrs are treated separately because they occur around hole boundaries; their displacement field is defined with respect to a detected circular rim and combines axial uplift, radial flare, tangential curl, and sparse tear-like protrusions. Online Table 2 summarizes these defect-specific displacement models, and Online Table 3 reports the hyperparameters used to instantiate them.